



GUIDELINE

Withdrawal Syndrome Management

Scope (Staff):	PCH medical, nursing, pharmacy and allied health staff
Scope (Area):	All PCH wards except NICU

Child Safe Organisation Statement of Commitment

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Aim

- To assist in the recognition, diagnosis, management and prevention of Iatrogenic Withdrawal Syndrome (IWS).
- To standardise the management of IWS across PCH.

Background

Opioids and benzodiazepines are regularly used in paediatric intensive care for sedation, analgesia and anxiolysis. Abrupt cessation or rapid tapering of opioids and/or benzodiazepines can lead to IWS.¹ This is characterised by autonomic dysregulation, neurologic symptoms, and gastrointestinal symptoms.¹ It occurs most commonly in infants and children who have received opioids and/or benzodiazepines for at least 5 days but can occur in those who have received as few as 3 days of treatment.^{1,2} The strongest risk factors are the total cumulative dose and the duration of treatment.^{1,3} Other risk factors include younger age (infants <6 months of age at greatest risk), pre-existing cognitive impairment and type of opioid medication, particularly fentanyl which is associated with a higher risk.^{3,4}

IWS has many overlapping symptoms with other conditions such as pain, distress, delirium and under-sedation (figure 1).⁵ These conditions should be excluded before the diagnosis of IWS can be confirmed.⁵

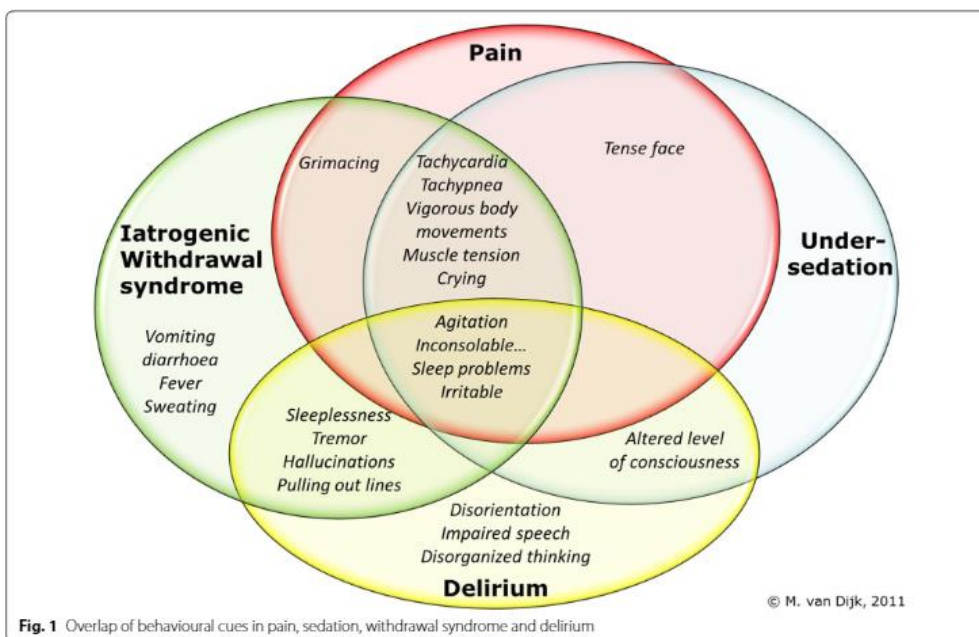


Fig. 1 Overlap of behavioural cues in pain, sedation, withdrawal syndrome and delirium

The Withdrawal Assessment Tool (WAT-1) is a validated tool for the assessment of withdrawal from opioids or benzodiazepines in infants and children and should be used to screen all patients who have received continuous infusions of opioids or benzodiazepines for ≥ 3 days. A WAT-1 score of 3 or greater suggests that the patient has IWS.⁵⁻⁸

Risk

Failing to anticipate, recognise or manage withdrawal can result in increased duration of mechanical ventilation, increased Intensive Care Unit (ICU) or hospital length of stay and patient and caregiver distress.^{1,2} Weaning opioids or benzodiazepines too slowly can result in over-sedation and it can also lead to prolonged ICU and hospital stay.¹ Higher cumulative exposure to opioids and benzodiazepines is associated with a potential negative impact on neurodevelopment.⁹

Key points

- Patients who receive IV infusions of opioids and / or benzodiazepines for 3 days or longer are at risk of IWS.
- The symptoms of IWS may overlap with the clinical signs of pain or distress, respiratory distress, delirium and noise induced stress. These other factors must be considered and excluded before the diagnosis can be confirmed.
- Patients who have received IV infusions of opioids and / or benzodiazepines for 5 days or longer should have their IV sedation converted to enteral doses and have these weaned down slowly depending on the risk of withdrawal. If enteral route is not established yet or there is poor oral absorption, then consider weaning sedation via IV route.
- Clonidine is commonly used in all patients weaning from opioids and /or benzodiazepines and is only weaned after the opioid and/or benzodiazepine are ceased.¹⁰⁻¹²
 - The rationale is that clonidine helps to treat the withdrawal symptoms of the other agents (particularly opioids), so it may facilitate a more rapid wean.
 - Many patients in the Paediatric Critical Care unit (PCC) would have received dexmedetomidine (which shares the same mechanism of action as clonidine) and the enteral clonidine may aid weaning from dexmedetomidine, although the exact conversions are unknown.
 - Clonidine is weaned over 3-5 days after the opioids and/or benzodiazepines have been ceased regardless of duration.
- WAT-1 scoring should be performed 12 hourly in all patients who are at risk of withdrawal. A WAT-1 score of 3 or higher is consistent with withdrawal and PRN medication should be considered if there is a clinical correlation.
- Withdrawal symptoms and signs can take up to 72 hours to manifest, even longer in renal failure. WAT-1 scoring should continue until 72 hours after the last dose of enteral opioid and/or benzodiazepine. Consider continuing WAT-1 scoring for a longer duration in patients with renal failure.
- Acute pain service (APS) should be involved in all patients transferred to a ward from PCC on a weaning regimen of enteral sedation.
- The process for weaning sedation is summarised in [Appendix 1: Withdrawal Algorithm](#).

Medical Management

Identification of Patients at Risk of IWS

All patients who have received intravenous infusions of opioids and/or benzodiazepines for 3 days or longer are at risk of IWS. There are 3 risk categories used at PCH:

1. Patients that received intravenous infusions of opioids and/or benzodiazepines for 3-4 days are considered low risk of withdrawal. These patients do not require conversion to enteral sedation but should be screened with WAT-1 scores at a minimum of 12 hourly for 48 hours post cessation of IV medications with PRN WAT-1 scores if clinical concerns of withdrawal.
2. Patients who have received IV infusions of opioids and/or benzodiazepines for 5-9 days are considered moderate risk of withdrawal. These patients should all be converted to enteral medication post extubation if possible (Refer to [Appendix 2: Opioid conversion](#) and [Appendix 3: Benzodiazepine conversion](#)) and these should be weaned by 20-25% of the original dose per day (i.e. a 4-5 day wean). Alternatively, if the patient cannot tolerate enteral medications for any reason, then the IV infusion should be continued and weaned by 20-25% of the original dose per day.
3. Patients who received IV infusions of opioids and/or benzodiazepines for ≥ 10 days are considered high risk of withdrawal. These patients should all be converted to enteral medication post extubation (Refer to [Appendix 2: Opioid conversion](#) and [Appendix 3: Benzodiazepine conversion](#)) and be weaned by 10-15% of original dose per day (i.e. an 8-10 day wean). Alternatively, if the patient cannot tolerate enteral medications for any reason, then the IV infusion should be continued and weaned by 10-15% of original dose per day.

Management Prior to Extubation

- Prior to extubation, titrate IV opioids to analgesia (e.g., morphine 10-20 microg/kg/hr or fentanyl 0.5-1 microg/kg/hr or halve the dose if the patient had been on high dose opioid infusion). Consider stopping IV benzodiazepines to allow the patient to wake sufficiently for extubation.
- Consider reducing IV benzodiazepines by 50% 12 hours prior to extubation.
- NB: This acute drop in sedation allows the patient to wake safely but may precipitate withdrawal. When calculating enteral conversion of opioids and benzodiazepines after extubation, it is important to use the dose that the patient was on prior to the dose reduction in preparation for extubation as a guide to calculating the enteral dose conversion.

Post Extubation

- After the patient is extubated then all patients at moderate or high risk of withdrawal (i.e., those who have received 5 days or more of opioids and/or benzodiazepines) will need to be commenced on regular enteral medication following the conversion doses in [Appendix 2: Opioid conversion](#) and [Appendix 3: Benzodiazepine conversion](#). If the patient cannot tolerate enteral medication for

any reason, then the IV sedation should be continued and weaned daily at as per the patient risk stratification (i.e., 20-25% daily for moderate risk and 10-15% daily for high-risk patients).

- Patients at moderate to high risk of withdrawal should also be commenced on regular clonidine, 1-2 microg/kg/dose enterally 6 hourly to aid in the management of withdrawal symptoms as per [Appendix 1: Withdrawal Algorithm](#). Refer to [Clonidine Monograph](#).
- It is important to stagger the doses of morphine, diazepam and clonidine to limit the risks of over-sedation and withdrawal. Ideally, these three medications should be spaced 2 hours apart, e.g.,
 - PO / NG morphine at 0200 / 0800 / 1400 / 2000
 - PO / NG diazepam 0400 / 1000 / 1600 / 2200
 - PO / NG clonidine at 0600 / 1200 / 1800 / 2400
- In children who are taking medication orally (i.e., those who do not have a nasogastric or gastrostomy tube in situ), consider clustering the medication earlier in the evening and later in the morning to allow an uninterrupted period of sleep while being mindful of the risk of over-sedation.

PRN Medications

- All patients identified as at risk of IWS should have PRN medications charted for use if WAT-1 score ≥ 3 . The following medications should be charted in order of use:
 - PO morphine 0.1 mg/kg 6 hourly (Refer to [Morphine Monograph](#))
 - PO Clonidine 1 microg/kg 6 hourly (Refer to [Clonidine Monograph](#))
 - PO Diazepam 0.1 mg/kg 6 hourly (Refer to [Diazepam Monograph](#))
- If a patient requires a dose of PRN medication, then a WAT-1 score should be repeated 60 minutes post the dose. A second or third line PRN medication should be considered if the WAT – 1 score remains ≥ 3 1 hour post PRN drug administration.

Weaning Medication

- Wean medication daily if the patient has required 2 or fewer doses of PRN medication in the last 24 hours.
- Wean enteral medication by 20-25% of original dose per day for patients at moderate risk of withdrawal.
- Wean enteral medications by 10-15% of original dose per day or 20-25% on alternate days for patients at high risk of withdrawal.
- It is important to use the original dose and not the current dose when weaning medication or else the weaning will not come to an end!

- Consider holding the weaning process for 24 hours if the patient has required >2 doses of PRN medication in the last 24 hours.
- If the patient reaches an absolute dose of 0.1 mg morphine or 0.1 mg diazepam, then wean by reducing the frequency of the dosing each day (i.e., from 6 hourly to 8 hourly) as it is not practical to administer smaller doses than this (a dilution would be required). Such small doses are often associated with 10-fold dosing errors.

Management of Withdrawal

- If a patient has a WAT-1 score ≥ 3 , then a medical review is required. Consider other causes for high WAT-1 scores initially. These include (but are not limited to) pain, discomfort, delirium, infection, seizures, anxiety, hypoxia, hypoglycaemia and electrolyte abnormalities.
- If no clear cause for the high WAT-1 score is identified, then the patient should have a PRN medication administered.
- The WAT-1 score should then be repeated 60 minutes after a PRN medication is given and if it remains ≥ 3 , then a second PRN medication should be administered and the WAT-1 score should be repeated again in 60 minutes.
- If there is another likely cause for the high WAT-1 score, such as patient requiring a nappy change or has had a recent painful procedure, then their pain or discomfort should be managed and the WAT-1 should be repeated in 30 minutes. If it continues to be ≥ 3 , then a PRN medication should be administered.
- If the patient is requiring multiple doses of a PRN medication to control a WAT-1 score, then consider increasing their regular dosing of morphine and/or diazepam. Alternatively, their regular clonidine dose may be increased by 1 microg/kg/dose to a maximum of 5 microg/kg 6 hourly including PRN doses (or absolute maximum of 200 microg/dose) to manage the symptoms of withdrawal. Refer to [Clonidine Monograph](#).
- All patients who have had a WAT-1 score of ≥ 3 should subsequently have WAT-1 scores done at least 4 hourly.

Discharge to Ward from PCC

- All patients at risk of withdrawal should have this clearly documented in the discharge plan on discharge from PCC.
- Patients at low risk of sedation should have the need for and duration of WAT-1 scoring documented in the discharge summary.
- For patients at moderate or high risk of withdrawal, it is important to include the original doses of morphine and/or diazepam in the discharge summary as well as the current weaning regimen (e.g., 10% or 20% daily reduction).
- The weaning doses for the following day post discharge should be charted and APS should be made aware of all patients prior to discharge from PCC.

Nursing Management

- Perform WAT-1 score 12 hourly on:
 - All patients on weaning enteral sedation until 48 hours after the weaning has ceased.
 - Patients who have received IV opioids / benzodiazepines for 3-4 days for 48 hours after cessation of IV opioid and benzodiazepines (PCC will advise of the need for and duration of WAT-1 scores when patients are discharged)
- WAT-1 should also be performed PRN if there are clinical concerns of withdrawal.
- If WAT-1 score is <3 , then the patient does not have withdrawal. Document this and continue to administer enteral sedation as charted.
- If WAT-1 score is ≥ 3 , then:
 1. Consider other causes for agitation. This includes pain, discomfort, anxiety, hypoxia, hypoglycaemia, infection and seizures.
 2. If another cause for agitation is found, then manage this and repeat WAT-1 score in 30-60 minutes.
 3. If WAT-1 score remains ≥ 3 , then treat patient for withdrawal with a PRN medication.
 4. Repeat WAT-1 score 1 hour after the PRN medication has been given. If the WAT-1 remains ≥ 3 , then a second PRN medication can be given and the WAT-1 is repeated in another hour.
 5. If the WAT-1 score remains ≥ 3 , then give a third PRN medication and notify APS.
 6. If a PRN medication is given within 1 hour of when the regular medication is due, then the regular medication should be delayed to allow an hour gap between the PRN and the regular medication and to prevent the risk of over-sedation.
 - For example, a patient is given a PRN dose of 0.1 mg/kg morphine and their regular morphine dose is due 30 minutes later. In this case the regular morphine dose should be delayed by 30 minutes to allow a 60-minute gap between the PRN and regular morphine and reduce the risk of over-sedation.
 - Waiting 30 minutes allows the PRN dose to reach their peak effect prior to giving the regular dose and thus reduce the risk of over-sedation.

Over-sedation

- If there are concerns that a patient is over-sedated (sleepy, not waking for feeds, low respiratory rate, apnoea) then escalate care to the medical team and withhold further doses of enteral sedatives (opioid, benzodiazepine or clonidine) until medical review.

- Refer to CPM Procedure [Resuscitation and Responding to Clinical Deterioration: MET Review and Code Blue](#)
- If the patient is clinically over-sedated, then consider holding or reducing the next prescribed dose of sedative.
- In cases of significant over-sedation, naloxone may be required. This is most likely to occur when the patient is first converted to enteral medication or if there is clustering of the doses of sedation before and after sleep.
- If the patient is in PCC at the time, then the PCC medical team is responsible for these changes.
- If the patient is on a general medical or surgical ward, then this should be discussed with APS.

References and related external legislation, policies, and guidelines
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12. Lee MM, Caylor K, Gockenback N. Evaluating the transition from dexmedetomidine to clonidine for the prevention of Withdrawal in Critically Ill Pediatric Patients. <i>J Pediatr Pharmacol Ther</i> . 2020;25(2):104-110
Related CAHS internal policies, procedures and guidelines
Morphine Monograph – Medication Management Manual
Clonidine Monograph – Medication Management Manual
Diazepam Monograph – Medication Management Manual
Resuscitation and Responding to Clinical Deterioration: MET Review and Code Blue (Clinical Practice Manual)
Useful resources
Bowens CD, Thompson JA, Thompson MT, Breitzka RL, Thompson DG, Sheeran PW, A trial of methadone tapering schedules in pediatric intensive care unit patients exposed to prolonged sedative infusions . <i>Journal of Paediatric Critical Care Medicine</i> . 2011; 12(5):504-511.
Children's Hospital of Philadelphia. PICU Pathway for Sedation/Analgesia in the Mechanically Ventilated Patient , 2016. Accessed May 2021

Queensland Children's Health. Clinical Guideline: Management of Analgesia, Sedation, Delirium, and Iatrogenic Withdrawal Syndrome in the PICU. Accessed May 2021. <i>(Not available online)</i>
Westmead Children's Hospital. Analgesia and Sedation PICU – CHW. Accessed May 2021. <i>(Not available online)</i>
Withdrawal Assessment Tool (WAT-1) (MR859.80)

This document can be made available in alternative formats on request for a person with a disability.

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Collaboration

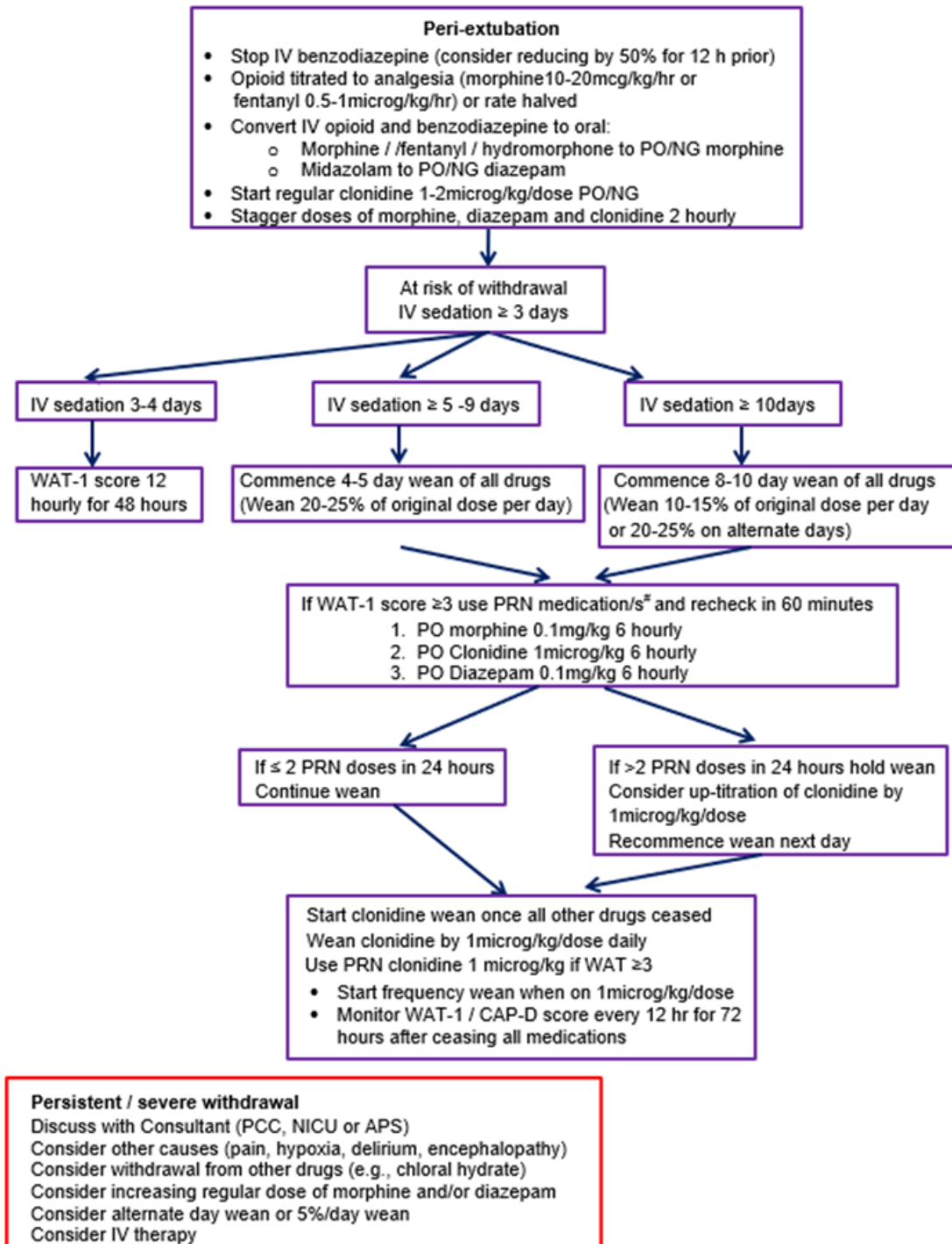
Accountability

Equity

Respect

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Appendix 1: Withdrawal Algorithm



Consider other causes of high WAT-1 score first, e.g., infection, pain / discomfort, delirium, hypoxia seizures, encephalopathy, electrolyte abnormality, hypoglycaemia

*Adapted from The Children's Hospital at Westmead, Queensland Children's Hospital and the Children's Hospital of Philadelphia

Appendix 2: Opioid Conversions*

• **Step 1:**

Calculate total daily opioid used in the 24 hours before the wean for extubation.

$\frac{\text{Total mL received in 24 hours} \times \text{concentration of infusion (mg/mL or microg/mL)}}{\text{Weight} \times 24 \text{ hours}}$

- Add up the total amount of infusion received in the 24 hours in mL from fluid chart.
- Calculate from this the total mg of opioid received in 24 hours based on the concentration of the infusion.
- Divide this by 24 and the patient weight to get the hourly opioid rate per kg including bolus doses.
- e.g., A 3 kg patient received 5 mL of 10 mg in 30 mL (0.33 mg/mL) IV morphine infusion over 24 hours. The total morphine received over 24 hours = 5 mL x 0.33 mg/mL = 1.65 mg. This equates to 1650 microg in 24 hours. To convert to a dose of infusion in microg/kg/hr divide by 24 x weight, i.e., = 1650 microg ÷ (24 hours x 3 kg) = 23 microg/kg/hr.

• **Step 2:**

If the patient was on fentanyl or hydromorphone infusion, then the next step is to convert this to a morphine equivalent using the following conversions:

- For fentanyl infusions multiply the hourly rate by 20
e.g., 1 microg/kg/hr fentanyl equates to 20 microg/kg/hr morphine.
- For hydromorphone infusions multiply the hourly rate by 3
e.g., 4 microg/kg/hr hydromorphone equates to 12 microg/kg/hr morphine.

• **Step 3:**

Finally use the following fixed conversions for morphine infusions to enteral morphine. Round to the nearest 10 microg/kg/hr dose.

- 10 microg/kg/hr = 0.1 mg/kg 6 hourly PO/NG morphine
- 20 microg/kg/hr = 0.2 mg/kg 6 hourly PO/NG morphine
- 30 microg/kg/hr = 0.25 mg/kg 6 hourly PO/NG morphine
- 40 microg/kg/hr = 0.3 mg/kg 6 hourly PO/NG morphine
- 50 microg/kg/hr = 0.35 mg/kg/hr 6 hourly PO/NG morphine
- 60 microg/kg/hr or higher = 0.4 mg/kg 6 hourly PO/NG morphine

• **Step 4:**

Chart the regular morphine dose 6 hourly and cease the opioid infusion 1 hour after the 1st dose of enteral morphine

**Adapted from The Children's Hospital at Westmead, Queensland Children's Hospital and the Children's Hospital of Philadelphia*

Appendix 3: Benzodiazepine Conversions*

mg/24 hours IV midazolam divided by 12 = mg 6 hourly PO/NG diazepam

- **Step 1:**

Calculate total daily IV midazolam dose including bolus doses.

- Add up the total volume of infusion received over the 24 hours prior to the wean from extubation from the fluid chart. Multiply this volume by the infusion concentration to obtain the total dose received in 24 hours.
- e.g., a 4 kg patient received 12 mL of 15 mg in 30 mL (0.5 mg/mL) midazolam infusion in the 24 hours prior to the wean for extubation. The total dose received is 12 mL x 0.5 mg/mL = 6 mg

- **Step 2:**

Convert IV midazolam dose to a total daily PO diazepam dose by dividing by 3.

- e.g., 6 mg IV midazolam = 2 mg PO diazepam

- **Step 3:**

Divide total daily PO diazepam dose by 4 to calculate a 6 hourly dose.

- e.g., 2 mg in 24 hours ÷ 4 = 0.5 mg PO 6 hourly

- **Step 4:**

Chart the 6 hourly PO diazepam dose as a regular dose on the medication chart. Cease the infusion of IV midazolam 1 hour after the 1st dose of PO diazepam.

**Adapted from The Children's Hospital at Westmead, Queensland Children's Hospital and the Children's Hospital of Philadelphia*